

FL Simulation Results

File: test_A5

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 80.0% (0.80)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 80.0% (0.80) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8743
Clean + DP Protection Accuracy: 0.8741
Poisoned Accuracy: 0.7992
Poisoned + DP Protection Accuracy: 0.7983
Drop (Clean - Poisoned): 0.0752
Drop (Clean - Init): 0.0815
Drop (Clean DP - Init): 0.0813
Drop (Poisoned - Init): 0.0064
Drop (Poisoned DP - Init): 0.0055
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8807
Clean Recall: 0.8743
Clean F1 Score: 0.8738

Clean DP Precision: 0.8807
Clean DP Recall: 0.8741
Clean DP F1 Score: 0.8735

Poisoned Precision: 0.8101
Poisoned Recall: 0.7992
Poisoned F1 Score: 0.7982

DP Protection Precision: 0.8095
DP Protection Recall: 0.7983
DP Protection F1 Score: 0.7974

Summary

FL Simulation Complete
Task: ddbf78f3-2fc8-4cb6-8b3a-4d6195321049
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8743
Clean DP Accuracy: 0.8741
Poisoned Accuracy: 0.7992
Data Poison Protection Accuracy: 0.7983
Drop (Clean - Poisoned): 0.0752
Drop (Clean - Init): 0.0815
Drop (Clean DP - Init): 0.0813
Drop (Poisoned - Init): 0.0064
Drop (Poisoned_DP - Init): 0.0055
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8807
Clean Recall: 0.8743
Clean F1 Score: 0.8738
Clean DP Precision: 0.8807
Clean DP Recall: 0.8741
Clean DP F1 Score: 0.8735
Poisoned Precision: 0.8101
Poisoned Recall: 0.7992
Poisoned F1 Score: 0.7982
DP Protection Precision: 0.8095
DP Protection Recall: 0.7983
DP Protection F1 Score: 0.7974